

The background of the slide is a light gray gradient. It is decorated with numerous realistic water droplets of various sizes. Some droplets are large and prominent, while others are small and subtle. They are scattered across the slide, with a higher concentration in the top-left and bottom-right corners, and a few in the center. Each droplet has a clear highlight and a soft shadow, giving it a three-dimensional appearance.

BIOMECHANICS OF WRIST

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Functions of the Wrist

- To argument fine hand and finger function
- To control length/ tension of finger long flexors and extensors
- To provide stability for skilled or forceful hand movements

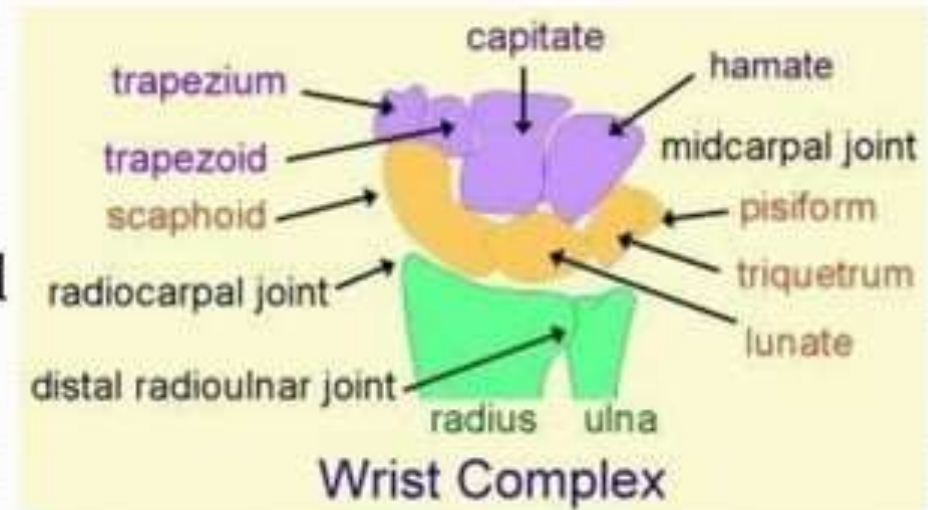
Joint Structure of the Wrist

- **Carpal bones**
- proximal row: loose arrangement

- scaphoid (navicular)
- Lunate
- Triquetrum
- Pisiform

- distal row: tightly bounded

- Trapezium
- trpezoid
- Capitate
- hamate



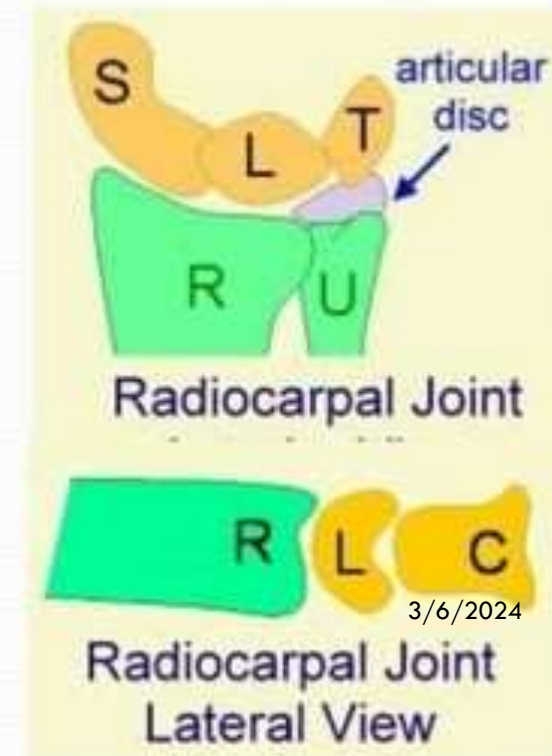
- NO muscles insert into the proximal carpal row

Joints at the wrist

- Radiocarpal joint:
 - primary joint for wrist extension
- Midcarpal joint :
 - primary joint for wrist flexion
- Intercarpal joints
- distal radioulnar joint :
 - forearm pronation/ supination

Radiocarpal joint

- **Proximal component:**
 - biconcave distal end of the radius and articular disc (TFCC)
 - Note : the carpal bones only articulate with the radius
- **Distal component:** biconvex proximal row of the carpal bones (scaphoid, lunate, and triquetrum)
- **Joint type:** condyloid, convex on concave



Cont...

- **Motions**

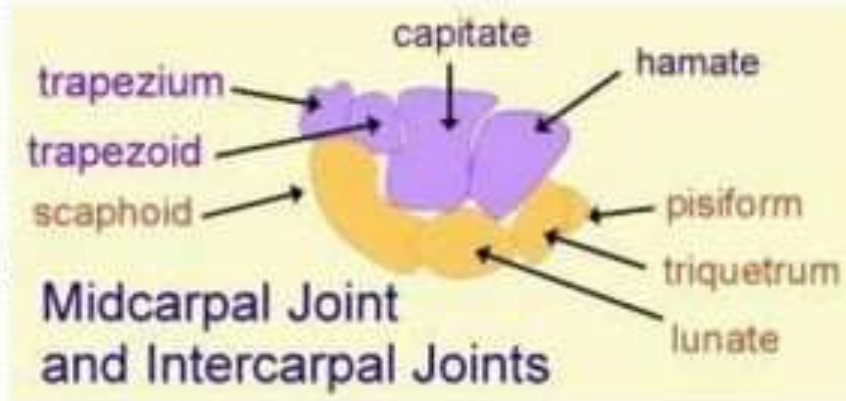
- wrist flexion/ extension with dorsal/ palmar glide of the proximal row of the carpal bones on the distal radius
- wrist radial/ ulnar deviation with ulnar/ radial glide of the proximal row of the carpal bones on the distal radius

- **DOF = 2**

- **Rest position** : slight wrist extension (10°)
- **Close-packed position** : full wrist extension

Midcarpal joint

- trapezoid and trapezium on scaphoid
- capitate on scaphoid
- capitate on lunate
- capitate on triquetrum
- hamate on triquetrum



Intercarpal joints

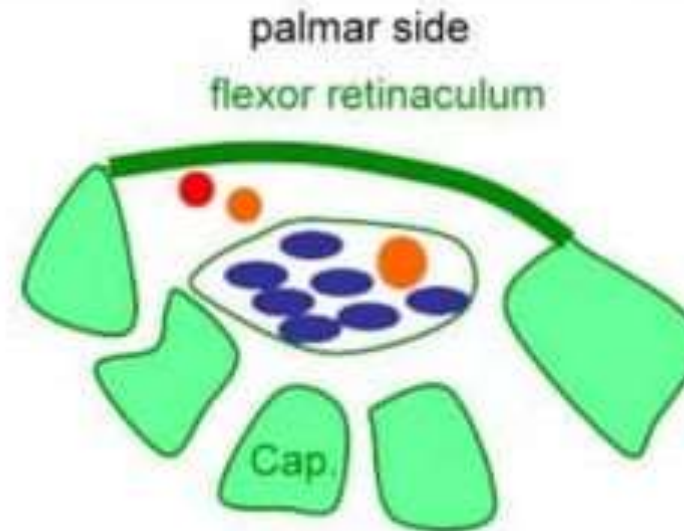
- trapezoid on capitate
- hamate on capitate
- pisiform on triquetrum

Distal radioulnar joint

- proximal component: convex ulnar head
- distal component: concave ulnar notch of the radius
- NOTE: different synovial capsule from the radiocarpal joint
- joint type: pivot joint
- Motions:
 - forearm pronation with anterior glide of the ulnar notch
 - forearm supination with posterior glide of the ulnar notch
- DOF = 1

Carpal tunnel

- proximal transverse arch covered with flexor retinaculum
- All extrinsic finger long flexors and median nerve passes through carpal tunnel except flexor digitorum ulnaris
- carpal tunnel syndrome
 - increase intratunnel pressure
 - compression of median nerve



Transverse View of Carpal Tunnel

3/6/2024

Anatomical snuff box

- also named forvea
- radialis area bounded by tendons of
 - extensor pollicis longus
 - extensor pollicis brevis

Muscles at the Wrist

- The axis of rotation for all wrist motions is located at the capitate; however, no wrist muscle actually crosses the wrist in line with this axis of rotation in any cardinal plane.
- All muscles create diagonal movements with varying moment arm to this axis of rotation.

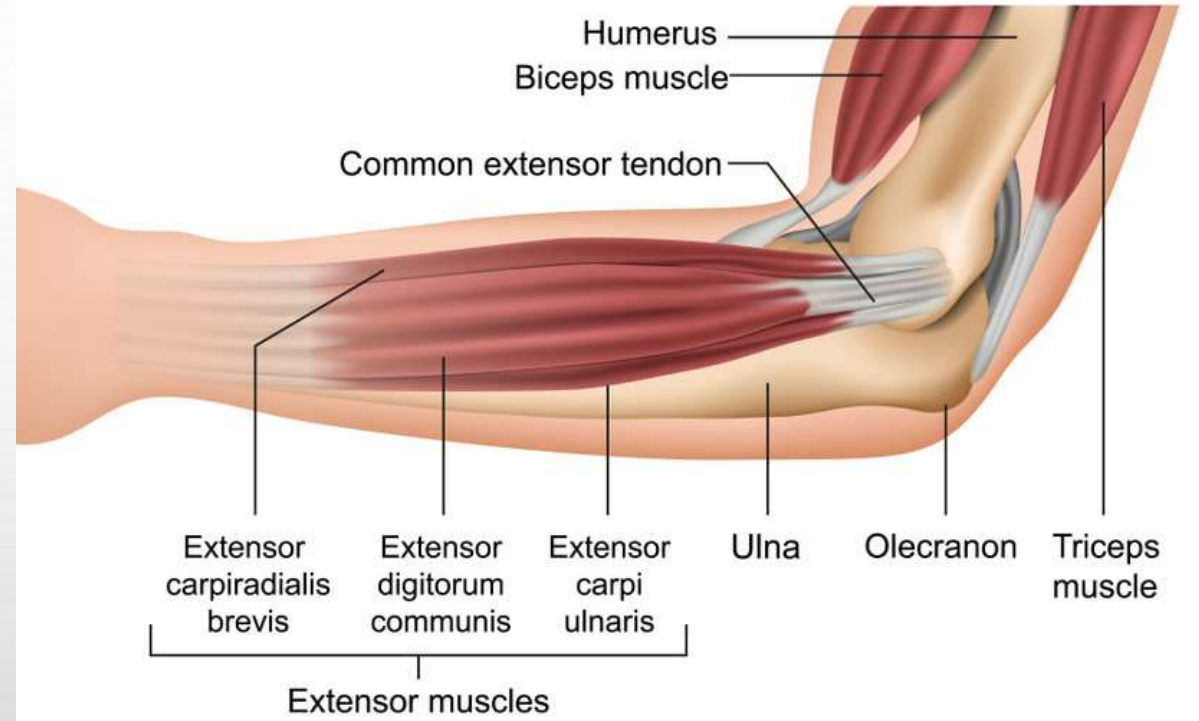
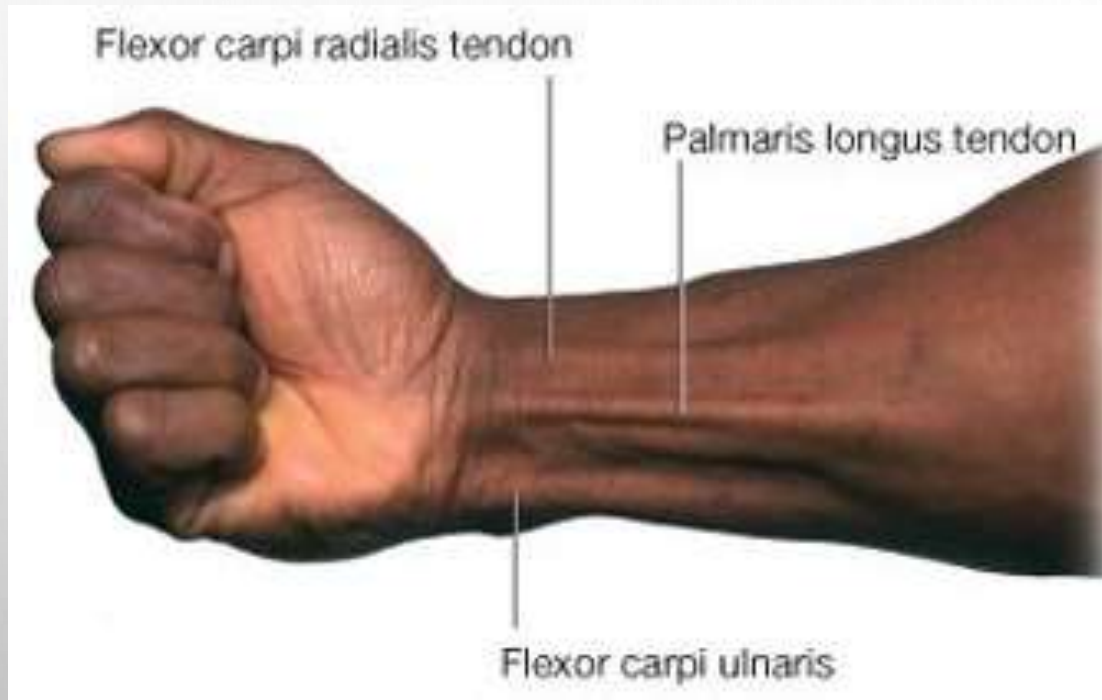
Cont...

- **Wrist flexors**

- Palmaris longus
- flexor carpi radialis
- flexor carpi ulnaris
- flexor digitorum superficialis

- **Wrist extensors**

- extensor carpi radialis
- longus extensor carpi radialis brevis
- extensor carpi ulnaris
- extensor digitorum communis



Cont...

- **Radial deviators**

- extensor carpi radialis longus : greatest as wrist in neutral
- extensor carpi radialis brevis
- flexor carpi radialis
- abductor pollicis longus
- extensor pollicis longus
- extensor pollicis brevis :
 - greatest moment arm but less cross-sectional area, resulting in less effective
- flexor pollicis longus

- **Ulnar deviators**

- flexor carpi ulnaris
- extensor carpi ulnaris
- radial : ulnar deviators = 1.15 : 1

Cont...

- **Forearm pronators**

- pronator teres
 - acts as forearm pronator
 - stabilizes proximal radioulnar joint
 - approximates humeroradial joint
- pronator quadratus
 - activates during all pronation activities

- **Forearm supinators**

- biceps brachii
- supinator

Kinematic

- **Wrist flexion and extension**

- Joint involved

- **Wrist flexion** (65 normal, 10 functional)

- radiocarpal joint: 40%
 - midcarpal joint: 60%

- **Wrist extension** (55 normal, 35 functional)

- radiocarpal joint: 67%
 - midcarpal joint: 33%

- **Plane of motion** : sagittal plane

- **Axis of rotation:**

- frontal axis through the center of the capitate
 - palpation : the area just proximal to the base of the third metacarpal bone



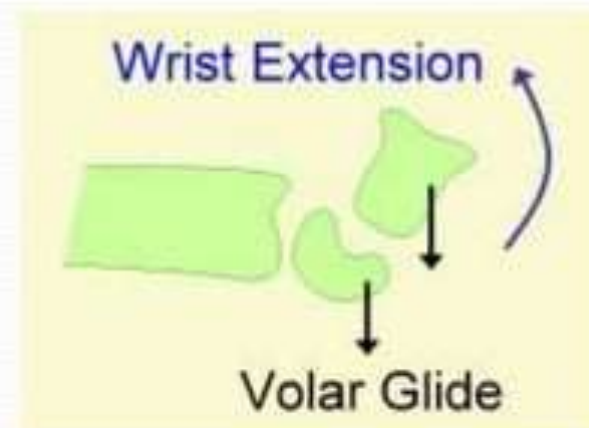
Osteokinematic movements

- Range of motion (ROM)
 - 65- 85° of wrist flexion
 - 55- 75° of wrist extension
- Functional range
 - from 10° of wrist flexion to 35° of wrist extension
- NOTE :
 - Loss of wrist function does not seriously impede performance of daily activities
- Closed kinematic chain motions
 - push-up
 - push against the wall

Arthrokinematic movements (convex on concave)

- Radiocarpal joint
 - distraction of the proximal row of the carpal bones on the radius
 - dorsal glide (posterior glide) of the proximal row of the carpal bones on the radius with wrist flexion
 - volar glide (palmar glide or anterior glide) of the proximal row of the carpal bones on the radius with wrist extension
- Midcarpal joint
 - distraction of the distal row of the carpal bones on the proximal row
 - dorsal glide (posterior glide) of the distal row of the carpal bones on the proximal row with wrist flexion
 - volar glide (palmar glide or anterior glide) of the distal row of the carpal bones on the proximal row with wrist extension
- Some intercarpal movements are also found during wrist motions

Cont...



Cont...

- factors limiting wrist flexion
 - dorsal radiocarpal ligaments:
 - radiolunate, radiotriquetrol, radioscapoid
- factors limiting wrist extension
 - dorsal surface of the radius
 - palmar ligaments : stronger than the dorsal ligaments

Cont...

- **Wrist radial/ ulnar deviation**

- **Joint involved**

- Radial deviation (abduction) (15-20 normal, 10 functional)

- most in midcarpal joint (15°)
 - radiocarpal joint
 - intercarpal joints

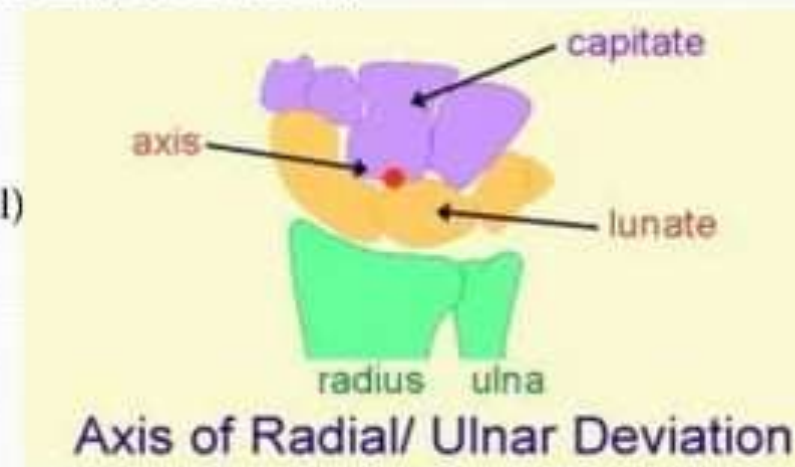
- Ulnar deviation (adduction)
(35-40 normal, 15 functional)

- most in radiocarpal joint (30°)
 - midcarpal joint
 - intercarpal joints

- **Plane of motion** : frontal plane

- **Axis of rotation**

- a line perpendicular to the plane of the palm through the intersection of the capitate and lunate
 - palpation : the area just proximal to the base of the third metacarpal bone is the capitate



Cont...

- Osteokinematic movements
 - Range of motion (ROM)
 - 0 ~ 15/ 25° of wrist radial deviation
 - 0 ~ 35/ 45° of wrist ulnar deviation
 - ulnar deviation > radial deviation because of ulnar tilt
 - Closed kinematic chain motions
 - to open or close a can

Arthrokinematic movements (convex on concave)

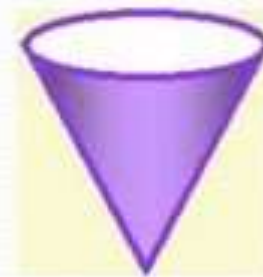
- Radiocarpal joint
 - ulnar glide of the proximal row of the carpal bones on the radius with wrist radial deviation
 - radial glide of the proximal row of the carpal bones on the radius with wrist ulnar deviation
- Midcarpal joint
 - ulnar glide of the distal row of the carpal bones on the proximal row with wrist radial deviation
 - radial glide of the distal row of the carpal bones on the proximal row with wrist ulnar deviation
- Some intercarpal movements are also found during wrist motions

Cont...

- Factors checking radial deviation
 - radial styloid process
 - ulnar collateral ligament : from styloid process of the ulna to pisiform and triquetrum
- Factors checking ulnar deviation
 - radial collateral ligament: from styloid process of the radius to scaphoid

Wrist Circumduction

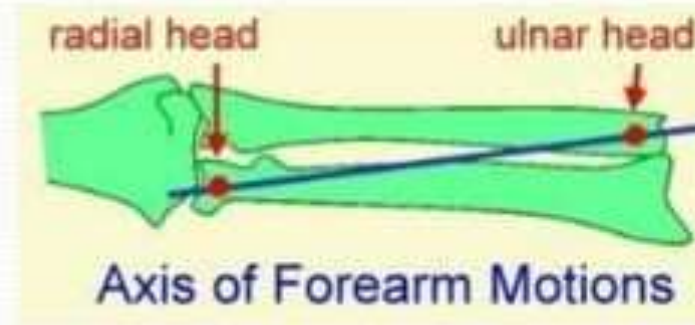
- cone-like motion that uses a combination of wrist flexion, extension, radial deviation, and ulnar deviation
- wrist flexion → radial deviation → extension
→ ulnar deviation
- wrist flexion → ulnar deviation → extension
→ radial deviation



Forearm pronation/ supination

- Joint involved

- Proximal radioulnar joint
- distal radioulnar joint
- humeroradial joint



- Plane of motion : transverse plane

- Axis of rotation

- passing through the centers of both the radial and the ulnar heads
- not parallel to the longitudinal axis of the forearm

Osteokinematic movements

- **Motions**

- the radial head rotates on the ulna within the annular ligament
- the radial head spins on the capitulum of the humerus
- the ulnar notch of the radius rotates on the ulnar head

- **Range of motion (ROM)**

- pronation : $0-70^{\circ}$
- supination : $0-85^{\circ}$
- Note: Pure motions occur as the elbow is held in the 90° of flexion.
- If the elbow is extended, shoulder rotation occurs simultaneously with forearm rotation.

- **Functional range** : 50° of pronation to 50° of supination

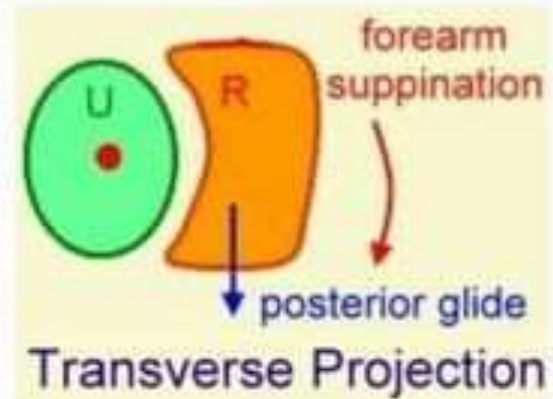
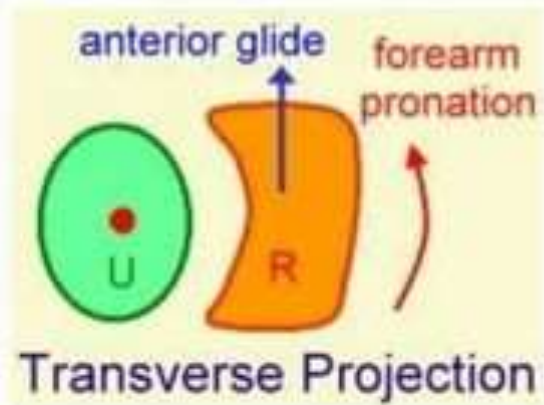
- **Closed kinematic chain motions**

- open the door knob
- open a can

Arthrokinematic movements

- Proximal radioulnar joint (convex on concave)
 - Anteromedial gliding of the radius on the ulna with forearm supination
 - Posterolateral gliding of the radius on the ulna with forearm pronation
- Humeroradial joint (concave on convex)
 - spinning of the radius on the humerus with forearm supination/ pronation
- Distal radioulnar joint (concave on convex)
 - anterior gliding of the radius on the ulna with forearm pronation
 - posterior gliding of the radius on the ulna with forearm supination

Cont...



- The ligaments of the wrist include
 - extrinsic ligaments
 - bridge carpal bones to the radius or metacarpals
 - include volar and dorsal ligaments
 - intrinsic ligaments
 - originate and insert on carpal bones
 - the most important intrinsic ligaments are the scapholunate interosseous ligament and lunotriquetral interosseous ligament
- Characteristics
 - volar ligaments are secondary stabilizers of the scaphoid
 - volar ligaments are stronger than dorsal ligaments
 - dorsal ligaments converge on the triquetrum

EXTRINSIC LIGAMENTS

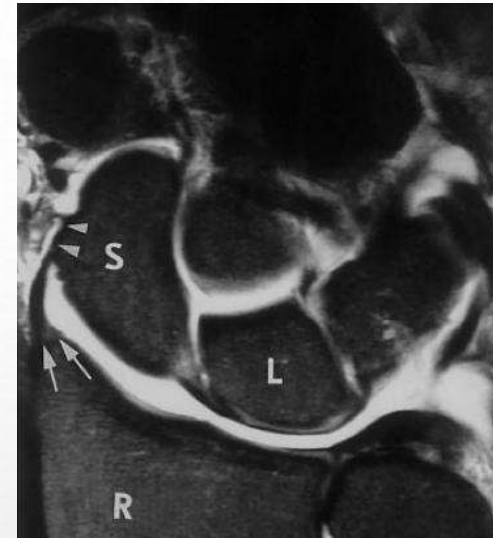
- VOLAR RADIOCARPAL LIGAMENTS

1. RADIAL COLLATERAL

2. RADIOSCAPHOCAPITATE

- AT RISK FOR INJURY WITH EXCESSIVELY LARGE RADIAL STYLOID

- FROM RADIAL STYLOID TO CAPITATE, CREATING A SLING TO SUPPORT THE WAIST OF THE SCAPHOID



3. LONG RADIOLUNATE

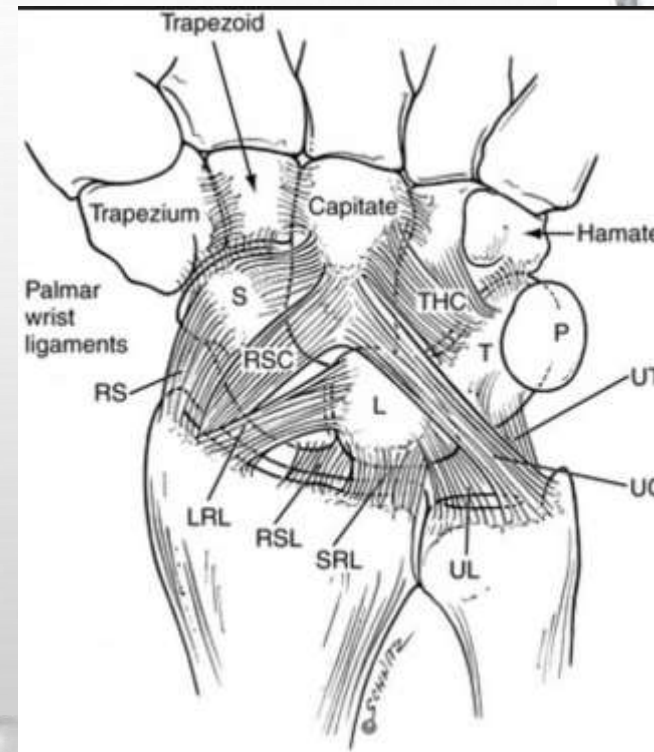
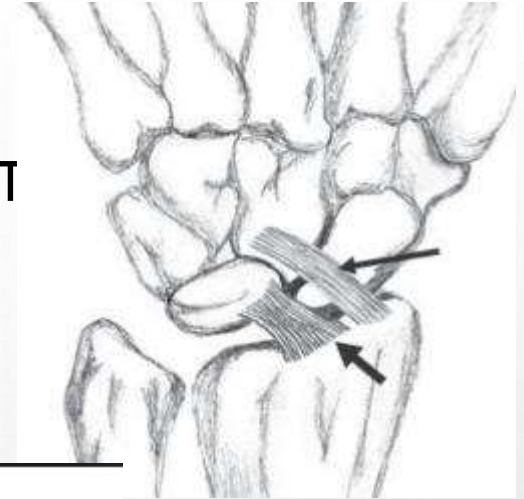
- COUNTERACTS ULNAR-DISTAL TRANSLOCATION OF THE LUNATE

4. RADIOSCAPHOLUNATE

- ONLY FUNCTIONS AS NEUROVASCULAR CONDUIT
- NOT A TRUE LIGAMENT
- DOES NOT ADD MECHANICAL STRENGTH

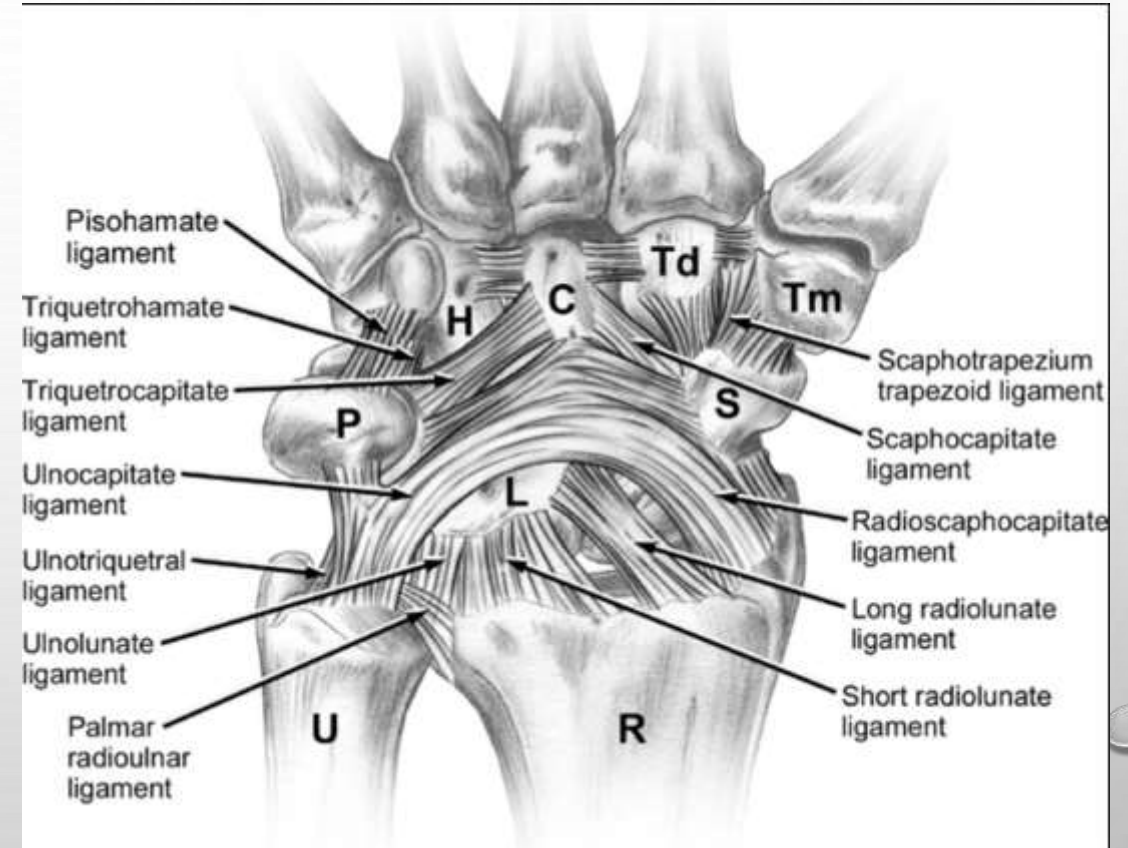
5. SHORT RADIOLUNATE

- STABILIZES LUNATE



VOLAR ULNOCARPAL LIGAMENTS

- ULNOTRIQUETRAL
- ULNOLUNATE
- ULNOCAPITATE



DORSAL LIGAMENTS

1. RADIOTRIQUETRAL

- ALSO REFERRED TO AS DORSAL RADIOCARPAL LIGAMENT (DRC)

2. DORSAL INTERCARPAL (DIC)

3. RADIOLUNATE

4. RADIOSCAPHOID

INTRINSIC (INTEROSSEOUS) LIGAMENTS

- PROXIMAL ROW

SCAPHOLUNATE LIGAMENT

- PRIMARY STABILIZER OF SCAPHOLUNATE JOINT

LUNOTRIQUETRAL LIGAMENT



DISTAL ROW

- TRAPEZIOTRAPEZOID LIGAMENT
- TRAPEZIOCAPITATE LIGAMENT
- CAPITOHAMATE LIGAMENT
-

SPACE OF POIRIER

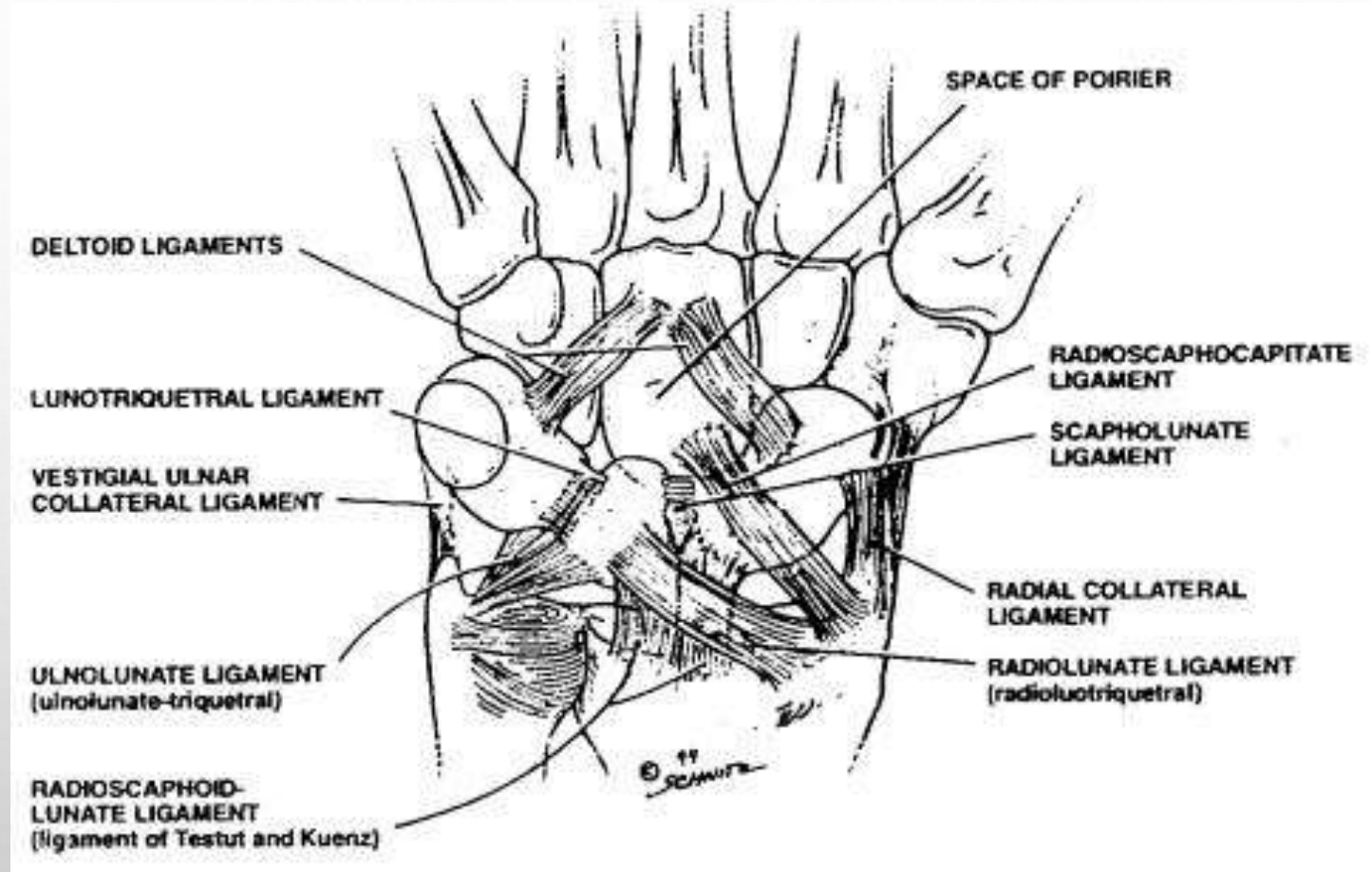
- CENTER OF A DOUBLE "V" SHAPE CONVERGENCE OF LIGAMENTS
- CENTRAL WEAK AREA OF THE WRIST IN THE FLOOR OF THE CARPAL TUNNEL AT THE LEVEL OF THE PROXIMAL CAPITATE
- BETWEEN THE VOLAR RADIOSCAPHOCAPITATE LIGAMENT AND VOLAR LONG RADIOLUNATE LIGAMENT (RADIOLUNOTRIQUETRAL LIGAMENT)

WRIST PALMAR FLEXION

- AREA OF WEAKNESS DISAPPEARS

WRIST DORSIFLEXION

- AREA OF WEAKNESS INCREASES



Wrist stability

- Bony configuration : most important
 - tension of ligaments
 - extrinsic and intrinsic ligaments
 - anterior and posterior radiocarpal ligaments
 - radial collateral ligament : prevents ulnar deviation stress
 - ulnar collateral ligament : prevents radial deviation stress
 - interosseous membrane
 - palmar carpal ligament (flexor retinaculum): to stabilize the tendons of the wrist flexors and prevent excessive bowstringing during flexion
- muscular arrangement
 - no muscle inserts on carpals excepts flexor carpi ulnaris

Stability of the distal radioulnar joint

- bony configuration
- interosseous membrane

Common Injury at the Wrist

- **Direct stress**
- compression stress → Colles' fracture (distal radius fracture) with/ without dorsal displacement
- **Repeated stresses**
- repeated wrist flexion/extension motion → chronic tenosynovitis of common flexor tendon → carpal tunnel syndrome
- repeated radial/ulnar deviation motion → chronic tenosynovitis of extensor pollicis brevis and/or abductor pollicis longus → deQuervain syndrome